

GRADUATE RESEARCH MANAGEMENT SYSTEM (GRMS)

FEDERAL URDU UNIVERSITY OF ARTS, SCIENCE & TECHNOLOGY

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PRESENTATION AGENDA

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INTRODUCTION

What is GRMS?

GRMS is a modern, web-based Graduate Research Management System custom-engineered for **Federal Urdu University of Arts, Science & Technology**.

Core Goal: To digitize and streamline the manual academic research workflows:

Synopsis Submission

Extension Cases

Semester Progress

Degree Letters

Graduate Research Concept



PROBLEM STATEMENT

The current manual framework at FUUAST struggles with severe operational bottlenecks:



Manual Paper Trails & Inefficiency

Synopsis and thesis submissions rely strictly on physical hard-copies, resulting in slow transit and easy losses.



Inefficient Progress Monitoring

No central dashboard exists to view research statuses. Coordinating changes between supervisors and students lacks clear transparency.



Manual Verification & Report Generation

Examiner evaluations, viva status recordings, and degree letter verifications are manually generated without systemic logging.

PURPOSE OF SYSTEM

To deliver a robust web portal that transforms FUUAST research workflow digitally:



End-to-End Automation

Entirely automates workflows from submission to approval cycles, reducing typical operational lead times by over 70%.



Integrated Role-Based Workspaces

Provides highly secure, distinct dashboards customized for Admin, Student, Supervisor, and External/Internal Examiners.



Real-time Alerts & Traceability

Maintains persistent activity trails, automated document versioning, and immediate visual page notifications for all actions.

CORE OBJECTIVES

1. SECURE CLOUD DELIVERY

Develop a highly available Django 5.x application complete with secure password hashing and visual responsiveness for mobile & desktop layouts.

2. PRECISE ACCESS MATRIX

Implement rigorous permission-based control lists, isolating students from dashboard operations of examiners and coordinators.

3. AUTOMATED DOCUMENT ENGINE

Provide secure file storage workflows supporting validation, size check mechanisms, and automated clean PDF degree letter exports.

4. FAST COLLABORATION

Facilitate synchronous or scheduled feedback cycles via scheduled evaluation boards, direct tasks, and prompt in-app messages.

PROPOSED ARCHITECTURE

Full-Scale Enterprise Web Portal

-  **Centralized DB Repository:** Normalised storage to keep track of every document and grade.
-  **Iterative Submission Loops:** Interactive feedback steps (Changes Requested → Resubmit).
-  **Cloud Hosting ready:** Live deployment using PythonAnywhere for quick validation.

The deadline for the submission of abstracts is extended to **11.59 pm** on **8th August, 2025**.

Call for Abstracts

The Department of Humanities and Social Sciences (HSS) at IIT Palakkad invites abstracts from PhD scholars enrolled in Indian universities/ institutes for the first edition of the Graduate Research Meet (GRM). The GRM is a PhD student-led event for young scholars from diverse disciplines in the humanities and social sciences to present their innovative research. It is envisioned as a platform for interdisciplinary dialogue and engagement with emerging research and methodologies.

An indicative but non-exhaustive list of themes is as follows:

- *Economics*: Economic Growth and Development
- *English*: Cultural Studies
- *History*: History of Modern South Asia, Environmental History, Agrarian History
- *Linguistics*: Phonetics, Sociolinguistics, Bilingual Language Acquisition, World Englishes
- *Philosophy*: Phenomenology and existentialism, Philosophy of technology, Continental Philosophy, Indian Philosophy
- *Political Science*: Social Policy Analysis, Urban Policy, Housing Studies
- *Psychology*: Sexualities, Genders, Relationship diversities and disabilities, Community Mental Health
- *Sociology*: Neoliberal economy and rural social change, Sociology of political change, Caste and Public Policy



PLATFORM USER ROLES (4 ROLES)

ROLE	DASHBOARD SCOPE	AUTHORIZED OPERATIONAL CONTROLS
 Admin	Central Management	User creation, examiner assignments, parameter controls, global system statistics.
 Student	Research Lifecycle	Upload synopsis/thesis files, progress reports, extension requests, download degree letters.
 Supervisor	Advisory Workspace	Approval reviews, version comparison, meeting scheduling, progress tracking metrics.
 Examiner	Evaluations Panel	Grade assigned files, enter scorecards, upload independent evaluation notes & reports.



FUNCTIONAL MODULES – PART 1



1. AUTHENTICATION

Built-in secure validation system featuring hashed password stores, active sessions log, and redirection to mapped user views.



2. ADMIN DASHBOARD

High-level administrative metrics panel supporting student/faculty CRUD actions, submission tracking, and platform settings.



3. STUDENT TERMINAL

Clean dashboard with PDF submission portal (5MB limit validation), direct tracking bars, and notifications bell indicator.



FUNCTIONAL MODULES – PART 2



4. SUPERVISOR CONTROL WORKSPACE

Dedicated interface designed specifically for review coordination and systematic student mapping:

- ✓ Complete synopsis & thesis reviewing interface
- 📅 Schedule physical or online virtual meetings
- 🕒 Endorse duration extensions with feedback notes



5. EXAMINER EVALUATION DESK

Enables highly focused thesis evaluations without student-side interference:

- 📄 Direct source file downloads
- ✓ Submit quantitative marks, grades, and written recommendations
- 🕒 Access evaluation history across cycles



FUNCTIONAL MODULES – PART 3



6. SYNOPSIS WORKFLOW

- Rigid PDF restriction
- Lifecycle: Draft → Submitted → Changes Requested → Approved
- Tracks resubmissions



7. THESIS WORKFLOW

- Internal & External examiner assignment system
- Scores and recommendations recording
- Viva results logging



8. PROGRESS LOGS

- Semester-wise reporting
- Supervisor response feedback
- Needs-Improvement flags



FUNCTIONAL MODULES – PART 4



9. MEETINGS TRACKER

Schedule research reviews (mode, date/time parameters) and log structured meeting minutes.



10. EXTENSION CASES

Process extension requests. Tracks pending cases with flags for urgent attention.



11. DEGREE LETTERS

Automated verification workflow generating printable PDF certificates via ReportLab.



CORE TECHNOLOGY STACK

LAYER	TECHNOLOGIES USED	PURPOSE / ROLE
Backend Core	Django 5.2 (Python)	Server logic, safe queries, admin routing
Frontend Stack	HTML5, CSS3, JavaScript, Bootstrap 5, AOS Library	Responsive interface, smooth page transitions
Database	MySQL	Relational persistent storage, 15+ tables
Security/PDF	Django Cryptography, ReportLab PDF Engine	Password safety & auto-generated certificates
Cloud Platform	PythonAnywhere	Production deployment & media file management



DATABASE SCHEMA & STRUCTURE



ESSENTIAL TABLES (15+ TOTAL)

accounts_student / accounts_supervisor
accounts_synopsis (Synopsis files & lifecycle states)
accounts_thesis / accounts_thesisevaluation
accounts_progressreport (Semester updates)
accounts_meeting / accounts_degreeletter



MODEL RELATIONSHIPS

One-to-One Relationships: Maps student and supervisor profiles securely directly to Django base User models.

Normalization: Structured in 3rd Normal Form (3NF) to minimize data redundancy and maximize query performance.

LIVE DEMO WORKFLOWS

01**Admin Controls**

Access global stats dashboard, add profiles, and assign academic committees.

02**Student Upload**

Submit research synopsis documents in safe PDF format and monitor status indicators.

03**Faculty Review**

Review proposals, request alterations, grade submissions, and verify results.

Live App URL: alishba154.pythonanywhere.com



FUTURE ENHANCEMENTS



AUTO SMTP ALERTS

Trigger automated emails to supervisors and students on review milestones.



NATIVE MOBILE APP

Build React Native apps for Android and iOS devices.



GOOGLE MEET & CHAT

Direct APIs to generate calendar events and provide real-time messaging.

✔ PROJECT CONCLUSION

✔ Successful Digital Transformation

GRMS replaces slow, manual paper tasks with clean automated workflows for graduate researchers at FUUAST.

✔ Higher Academic Efficiency

Improves transparency, prevents documentation losses, and automates degree verification letter exports.

✔ Production Tested Deployment

Fully configured with custom responsive user views and live deployed on cloud infrastructure.

THANK YOU!

Do you have any questions or feedback?



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University: FUUAST Computer Science



alishba154.pythonanywhere.com



github.com/alishba158/GRMS

IMAGE SOURCES



<https://greyai.ca/assets/dashboard-mockup.png>

Source: greyai.ca



<https://iitpkd.ac.in/sites/default/files/inline-images/Abstracts-%20Submission%20dates%20extended.%20.jpg>

Source: iitpkd.ac.in